```
superxoma@superxoma:~/devops11kube$ kubectl get deployment
NAME        READY   UP-TO-DATE   AVAILABLE   AGE
flask-app   5/5     5            5           10d
redis       1/1     1            1           10d
```

```
superxoma@superxoma:~/devops11kube$ kubectl delete all --all
pod "flask-app-78f58b95c5-6rjns" deleted from default namespace
pod "flask-app-78f58b95c5-6spqn" deleted from default namespace
pod "flask-app-78f58b95c5-9vfwn" deleted from default namespace
pod "flask-app-78f58b95c5-r5fzr" deleted from default namespace
pod "flask-app-78f58b95c5-r8m5k" deleted from default namespace
pod "redis-9f7d5587-575vz" deleted from default namespace
service "kubernetes" deleted from default namespace
service "redis" deleted from default namespace
service "service-devops" deleted from default namespace
deployment.apps "flask-app" deleted from default namespace
deployment.apps "redis" deleted from default namespace
```

```
superxoma@superxoma:~/devops11kube$ curl -fsSL -o get_helm.sh https://raw.githubusercontent.com/helm/helm/main/scripts/g
et-helm-3
superxoma@superxoma:~/devops11kube$ chmod 700 get_helm.sh
superxoma@superxoma:~/devops11kube$ ./get_helm.sh
Downloading https://get.helm.sh/helm-v3.20.1-linux-amd64.tar.gz
Verifying checksum... Done.
Preparing to install helm into /usr/local/bin
helm installed into /usr/local/bin/helm
```

```
superxoma@superxoma:~/devops11kube$ helm create PromGrafana
Creating PromGrafana
superxoma@superxoma:~/devops11kube$ tree PromGrafana/
PromGrafana/
├── charts
├── Chart.yaml
├── templates
│   ├── deployment.yaml
│   ├── _helpers.tpl
│   ├── hpa.yaml
│   ├── httproute.yaml
│   ├── ingress.yaml
│   ├── NOTES.txt
│   ├── serviceaccount.yaml
│   ├── service.yaml
│   └── tests
│       └── test-connection.yaml
└── values.yaml

4 directories, 11 files
```

```
superxoma@superxoma:~/devops11kube$ curl -L https://github.com/kubernetes/kompose/releases/download/v1.34.0/kompose-linu
x-amd64 -o kompose
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
100 21.0M  100 21.0M    0     0  1142k      0  0:00:18  0:00:18 --:--:-- 1783k
superxoma@superxoma:~/devops11kube$ chmod +x kompose
superxoma@superxoma:~/devops11kube$ sudo mv ./kompose /usr/local/bin/kompose
```

```
superxoma@superxoma:~/devops11kube/devops9prom$ kompose convert
WARN Restart policy 'unless-stopped' in service prometheus is not supported, convert it to 'always'
WARN Restart policy 'unless-stopped' in service grafana is not supported, convert it to 'always'
WARN File don't exist or failed to check if the directory is empty: stat :/var/lib/grafana: no such file or directory
WARN File don't exist or failed to check if the directory is empty: stat :/prometheus: no such file or directory
INFO Kubernetes file "blackbox-service.yaml" created
INFO Kubernetes file "grafana-service.yaml" created
INFO Kubernetes file "prometheus-service.yaml" created
INFO Kubernetes file "blackbox-deployment.yaml" created
INFO Kubernetes file "grafana-deployment.yaml" created
INFO Kubernetes file "grafana-data-persistentvolumeclaim.yaml" created
INFO Kubernetes file "grafana-cm0-configmap.yaml" created
INFO Kubernetes file "prometheus-deployment.yaml" created
INFO Kubernetes file "prom-data-persistentvolumeclaim.yaml" created
INFO Kubernetes file "prometheus-cm0-configmap.yaml" created
```

```
superxoma@superxoma:~/devops11kube/devops9prom$ ls -l
total 52
-rw-r--r-- 1 superxoma superxoma  656 мар 30 07:38 blackbox-deployment.yaml
-rw-r--r-- 1 superxoma superxoma  306 мар 30 07:38 blackbox-service.yaml
-rw-rw-r-- 1 superxoma superxoma  756 мар 30 07:28 compose.yaml
drwxrwxr-x 2 superxoma superxoma 4096 мар 30 07:28 grafana
-rw-r--r-- 1 superxoma superxoma  310 мар 30 07:38 grafana-cm0-configmap.yaml
-rw-r--r-- 1 superxoma superxoma  210 мар 30 07:38 grafana-data-persistentvolumeclaim.yaml
-rw-r--r-- 1 superxoma superxoma 1222 мар 30 07:38 grafana-deployment.yaml
-rw-r--r-- 1 superxoma superxoma  303 мар 30 07:38 grafana-service.yaml
-rw-r--r-- 1 superxoma superxoma  204 мар 30 07:38 prom-data-persistentvolumeclaim.yaml
drwxrwxr-x 2 superxoma superxoma 4096 мар 30 07:28 prometheus
-rw-r--r-- 1 superxoma superxoma 1408 мар 30 07:38 prometheus-cm0-configmap.yaml
-rw-r--r-- 1 superxoma superxoma 1124 мар 30 07:38 prometheus-deployment.yaml
-rw-r--r-- 1 superxoma superxoma  312 мар 30 07:38 prometheus-service.yaml
```

```
superxoma@superxoma:~/devops11kube/devops9prom$ kompose convert --chart
WARN Restart policy 'unless-stopped' in service grafana is not supported, convert it to 'always'
WARN Restart policy 'unless-stopped' in service prometheus is not supported, convert it to 'always'
WARN File don't exist or failed to check if the directory is empty: stat :/var/lib/grafana: no such file or directory
WARN File don't exist or failed to check if the directory is empty: stat :/prometheus: no such file or directory
INFO Kubernetes file "compose/templates/blackbox-service.yaml" created
INFO Kubernetes file "compose/templates/grafana-service.yaml" created
INFO Kubernetes file "compose/templates/prometheus-service.yaml" created
INFO Kubernetes file "compose/templates/blackbox-deployment.yaml" created
INFO Kubernetes file "compose/templates/grafana-deployment.yaml" created
INFO Kubernetes file "compose/templates/grafana-data-persistentvolumeclaim.yaml" created
INFO Kubernetes file "compose/templates/grafana-cm0-configmap.yaml" created
INFO Kubernetes file "compose/templates/prometheus-deployment.yaml" created
INFO Kubernetes file "compose/templates/prom-data-persistentvolumeclaim.yaml" created
INFO Kubernetes file "compose/templates/prometheus-cm0-configmap.yaml" created
INFO chart created in "compose/"
```

```
superxoma@superxoma:~/devops11kube/devops9prom$ tree compose
compose
├── Chart.yaml
├── README.md
└── templates
    ├── blackbox-deployment.yaml
    ├── blackbox-service.yaml
    ├── grafana-cm0-configmap.yaml
    ├── grafana-data-persistentvolumeclaim.yaml
    ├── grafana-deployment.yaml
    ├── grafana-service.yaml
    ├── prom-data-persistentvolumeclaim.yaml
    ├── prometheus-cm0-configmap.yaml
    ├── prometheus-deployment.yaml
    └── prometheus-service.yaml
```

```
└── promgra
    ├── Chart.yaml
    ├── README.md
    └── templates
        ├── blackbox-deployment.yaml
        ├── blackbox-service.yaml
        ├── grafana-cm0-configmap.yaml
        ├── grafana-data-persistentvolumeclaim.yaml
        ├── grafana-deployment.yaml
        ├── grafana-service.yaml
        ├── prom-data-persistentvolumeclaim.yaml
        ├── prometheus-cm0-configmap.yaml
        ├── prometheus-deployment.yaml
        └── prometheus-service.yaml
```

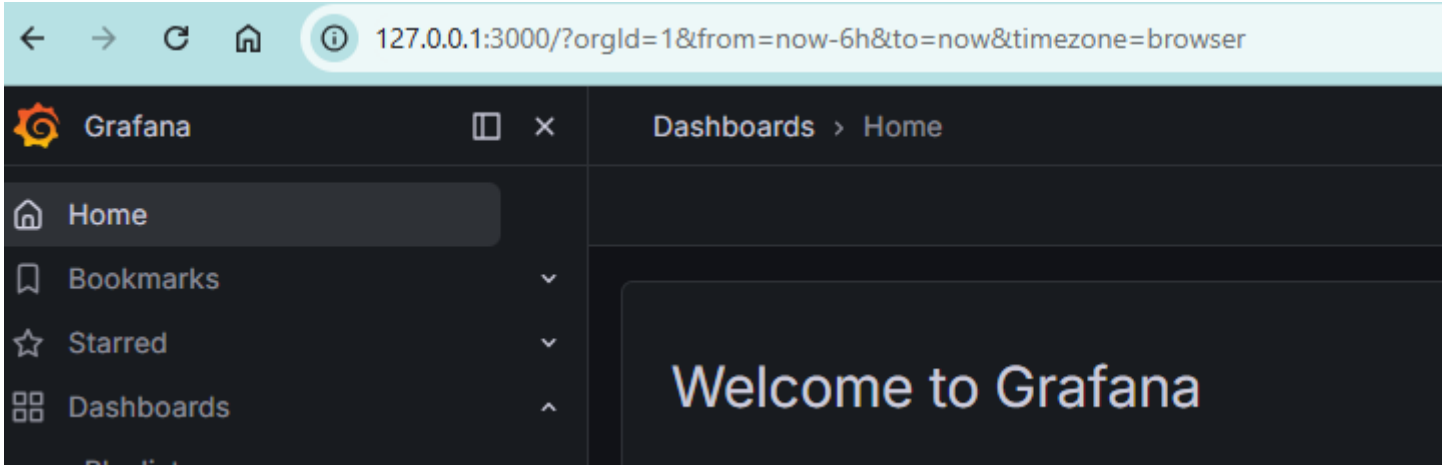
```
superxoma@superxoma:~/devops11kube/devops9prom$ helm package promgra
Successfully packaged chart and saved it to: /home/superxoma/devops11kube/devops9prom/promgra-0.0.1.tgz
```

```
superxoma@superxoma:~/devops11kube/devops9prom$ helm install promgra ./promgra-0.0.1.tgz
NAME: promgra
LAST DEPLOYED: Mon Mar 30 07:59:51 2026
NAMESPACE: default
STATUS: deployed
REVISION: 1
TEST SUITE: None
```

```
superxoma@superxoma:~/devops11kube/devops9prom$ kubectl get deployments
NAME         READY   UP-TO-DATE   AVAILABLE   AGE
blackbox     1/1     1            1           4m7s
grafana      1/1     1            1           4m7s
prometheus   1/1     1            1           4m7s
```

```
superxoma@superxoma:~/devops11kube/devops9prom$ kubectl get services
NAME         TYPE           CLUSTER-IP      EXTERNAL-IP           PORT(S)          AGE
blackbox     ClusterIP      10.98.143.215   <none>                9115/TCP         15m
grafana      LoadBalancer   10.107.244.85   127.0.0.1,10.0.2.15   3000:31320/TCP   15m
kubernetes   ClusterIP      10.96.0.1       <none>                443/TCP          51m
prometheus   ClusterIP      10.98.213.116   <none>                9090/TCP         15m
```

```
superxoma@superxoma:~/devops11kube/devops9prom$ kubectl get pods
NAME                              READY   STATUS    RESTARTS      AGE
blackbox-8564786cf9-t5lhk         1/1     Running   1 (12m ago)   16m
grafana-764577f489-b7mhv          1/1     Running   1 (12m ago)   16m
prometheus-6c45dc4766-cwzvq       1/1     Running   1 (12m ago)   16m
```




```
superxoma@superxoma:~$ minikube tunnel --bind-address 10.0.2.15
✅  Tunnel successfully started

📌  NOTE: Please do not close this terminal as this process must stay alive for the tunnel to be accessible ...

🏃  Starting tunnel for service grafana.
```

```
  GNU nano 7.2                                          values.yaml
EXTERNAL_IP: 10.0.2.15
EXTERNAL_PORT: 3113
GF_ADMIN_PASSWORD: HiGrafana
```

```
superxoma@superxoma:~/devops11kube/devops9prom/promgra$ helm lint
==> Linting .
[INFO] Chart.yaml: icon is recommended

1 chart(s) linted, 0 chart(s) failed
```

```
superxoma@superxoma:~/devops11kube/devops9prom$ helm show values ./promgra
EXTERNAL_IP: 10.0.2.15
EXTERNAL_PORT: 3113
GF_ADMIN_PASSWORD: HiGrafana
```

```
superxoma@superxoma:~/devops11kube/devops9prom$ helm template promgraf ./promgra
---
```

```
# Source: promgra/templates/grafana-service.yaml
apiVersion: v1
kind: Service
metadata:
  annotations:
    kompose.cmd: kompose convert --chart
    kompose.version: 1.34.0 (cbf2835db)
  labels:
    io.kompose.service: grafana
  name: grafana
spec:
  type: LoadBalancer
  ports:
    - name: "3000"
      port: 3000
      targetPort: 3000
  externalIPs:
    - 10.0.2.15
  selector:
    io.kompose.service: grafana
```

```
superxoma@superxoma:~/devops11kube/devops9prom/promgra$ kubectl get services
NAME          TYPE           CLUSTER-IP      EXTERNAL-IP           PORT(S)          AGE
blackbox      ClusterIP      10.98.143.215   <none>                9115/TCP         54m
grafana       LoadBalancer   10.107.244.85   127.0.0.1,10.0.2.15   3456:32025/TCP   54m
kubernetes    ClusterIP      10.96.0.1       <none>                443/TCP          90m
prometheus    ClusterIP      10.98.213.116   <none>                9090/TCP         54m
```

```
superxoma@superxoma:~/devops11kube/devops9prom/promgra$ helm get values promgra
USER-SUPPLIED VALUES:
EXTERNAL_PORT: 3456
```

```
superxoma@superxoma:~/devops11kube/devops9prom/promgra$ helm get values promgra --all
COMPUTED VALUES:
EXTERNAL_IP: 10.0.2.15
EXTERNAL_PORT: 3456
GF_ADMIN_PASSWORD: HiGrafana
```

127.0.0.1:3456/?orgId=1&from=now-6h&to=now&timezone=browser

Dashboards › Home

```
superxoma@superxoma:~/devops11kube/kustomize/base$ tree
.
├── flask_redis
│   ├── app.py
│   ├── dockerfile
│   └── requirements.txt
├── flask-service.yml
├── flask.yml
├── kustomization.yaml
├── redis-service.yml
└── redis.yml
```

```
superxoma@superxoma:~/devops11kube/kustomize$ kubectl kustomize base
apiVersion: v1
kind: Service
metadata:
  labels:
    app: devops-course-2025
  name: redis
spec:
  ports:
  - port: 6379
    targetPort: 6379
  selector:
    app: devops-course-2025
    svc: db
  type: ClusterIP
---
apiVersion: v1
kind: Service
metadata:
  labels:
    app: devops-course-2025
  name: service-devops
spec:
  externalIPs:
  - 10.0.2.15
  ports:
  - port: 8000
```

```
superxoma@superxoma:~/devops11kube/kustomize$ tree
.
├── base
│   ├── flask_redis
│   │   ├── app.py
│   │   ├── dockerfile
│   │   └── requirements.txt
│   ├── flask-service.yml
│   ├── flask.yml
│   ├── kustomization.yaml
│   ├── redis-service.yml
│   └── redis.yml
├── dev
│   └── kustomization.yaml
└── prod
    └── kustomization.yaml
```

```
superxoma@superxoma:~/devops11kube/kustomize$ kubectl kustomize dev
apiVersion: v1
kind: Service
metadata:
  labels:
    app: devops-course-2025
    environment: dev
  name: dev-redis
spec:
  ports:
  - port: 6379
    targetPort: 6379
  selector:
    app: devops-course-2025
    environment: dev
    svc: db
  type: ClusterIP
```

```
superxoma@superxoma:~/devops11kube/kustomize$ kubectl kustomize prod
apiVersion: v1
kind: Service
metadata:
  labels:
    app: devops-course-2025
    environment: prod
  name: prod-redis
spec:
  ports:
  - port: 6379
    targetPort: 6379
  selector:
    app: devops-course-2025
    environment: prod
    svc: db
  type: ClusterIP
```

```
superxoma@superxoma:~/devops11kube/kustomize$ tree
.
├── base
│   ├── flask_redis
│   │   ├── app.py
│   │   ├── dockerfile
│   │   └── requirements.txt
│   ├── flask-service.yml
│   ├── flask.yml
│   ├── kustomization.yaml
│   ├── redis-service.yml
│   └── redis.yml
├── dev
│   ├── kustomization.yaml
│   └── service-patch.yaml
└── prod
    └── kustomization.yaml
```

```
superxoma@superxoma:~/devops11kube/kustomize$ kubectl kustomize dev
```

```
spec:
  externalIPs:
  - 10.0.2.15
  ports:
  - port: 54321
    targetPort: 5000
```

```
superxoma@superxoma:~/devops11kube/kustomize$ kubectl apply -k dev/
service/dev-redis created
service/dev-service-devops created
deployment.apps/dev-flask-app created
deployment.apps/dev-redis created
```

```
superxoma@superxoma:~/devops11kube/kustomize$ kubectl get services
NAME                 TYPE           CLUSTER-IP      EXTERNAL-IP           PORT(S)           AGE
blackbox             ClusterIP      10.98.143.215   <none>                9115/TCP          100m
dev-redis            ClusterIP      10.102.28.134   <none>                6379/TCP          2m20s
dev-service-devops   LoadBalancer   10.96.30.55     127.0.0.1,10.0.2.15   54321:32638/TCP   2m20s
grafana              LoadBalancer   10.107.244.85   127.0.0.1,10.0.2.15   3456:32025/TCP    100m
kubernetes           ClusterIP      10.96.0.1       <none>                443/TCP           136m
prometheus           ClusterIP      10.98.213.116   <none>                9090/TCP          100m
```

```
superxoma@superxoma:~/devops11kube/kustomize$ kubectl get deployments
NAME            READY   UP-TO-DATE   AVAILABLE   AGE
blackbox        1/1     1            1           101m
dev-flask-app   5/5     5            5           3m18s
dev-redis       1/1     1            1           3m18s
grafana         1/1     1            1           101m
prometheus      1/1     1            1           101m
```

```
superxoma@superxoma:~/devops11kube/kustomize$ kubectl apply -k prod
service/prod-redis created
service/prod-service-devops created
deployment.apps/prod-flask-app created
deployment.apps/prod-redis created
superxoma@superxoma:~/devops11kube/kustomize$ kubectl get services
NAME                  TYPE           CLUSTER-IP      EXTERNAL-IP           PORT(S)           AGE
blackbox              ClusterIP      10.98.143.215   <none>                9115/TCP          102m
dev-redis             ClusterIP      10.102.28.134   <none>                6379/TCP          4m1s
dev-service-devops    LoadBalancer   10.96.30.55     127.0.0.1,10.0.2.15   54321:32638/TCP   4m1s
grafana               LoadBalancer   10.107.244.85   127.0.0.1,10.0.2.15   3456:32025/TCP    102m
kubernetes            ClusterIP      10.96.0.1       <none>                443/TCP           138m
prod-redis            ClusterIP      10.96.204.47    <none>                6379/TCP          6s
prod-service-devops   LoadBalancer   10.111.53.43    127.0.0.1,10.0.2.15   8000:32039/TCP    6s
prometheus            ClusterIP      10.98.213.116   <none>                9090/TCP          102m
superxoma@superxoma:~/devops11kube/kustomize$ kubectl get deployments
NAME            READY   UP-TO-DATE   AVAILABLE   AGE
blackbox        1/1     1            1           102m
dev-flask-app   5/5     5            5           4m10s
dev-redis       1/1     1            1           4m10s
grafana         1/1     1            1           102m
prod-flask-app  5/5     5            5           15s
prod-redis      1/1     1            1           15s
prometheus      1/1     1            1           102m
```

```
superxoma@superxoma:~/devops11kube/kustomize$ tree
.
├── base
│   ├── flask_redis
│   │   ├── app.py
│   │   ├── dockerfile
│   │   └── requirements.txt
│   ├── flask-service.yml
│   ├── flask.yml
│   ├── kustomization.yaml
│   ├── redis-service.yml
│   └── redis.yml
├── dev
│   ├── deployment-patch.yaml
│   ├── kustomization.yaml
│   └── service-patch.yaml
└── prod
    └── kustomization.yaml
```

```
superxoma@superxoma:~/devops11kube/kustomize$ kubectl apply -k dev/
service/dev-redis unchanged
service/dev-service-devops unchanged
deployment.apps/dev-flask-app configured
deployment.apps/dev-redis unchanged
superxoma@superxoma:~/devops11kube/kustomize$ kubectl get pods
NAME                              READY   STATUS    RESTARTS       AGE
blackbox-8564786cf9-t5lhk         1/1     Running   1 (101m ago)   105m
dev-flask-app-778b87956-j6nqs     1/1     Running   0              7m14s
dev-flask-app-778b87956-n5bp8     1/1     Running   0              7m14s
dev-redis-684ff98885-2qlzk        1/1     Running   0              7m14s
grafana-764577f489-b7mhv          1/1     Running   1 (101m ago)   105m
prod-flask-app-68dbccb746-4hvgx   1/1     Running   0              3m19s
prod-flask-app-68dbccb746-77h7f   1/1     Running   0              3m19s
prod-flask-app-68dbccb746-l9tzs   1/1     Running   0              3m19s
prod-flask-app-68dbccb746-pmdrp   1/1     Running   0              3m19s
prod-flask-app-68dbccb746-stmhs   1/1     Running   0              3m19s
prod-redis-74477d6b7f-qk4nh       1/1     Running   0              3m19s
prometheus-6c45dc4766-cwzvq       1/1     Running   1 (101m ago)   105m
```

```
superxoma@superxoma:~/devops11kube/kustomize$ minikube image build -t flask:v2 base/flask_redis/
#0 building with "default" instance using docker driver

#1 [internal] load build definition from dockerfile
#1 transferring dockerfile: 337B 0.0s done
#1 DONE 0.2s

#2 resolve image config for docker-image://docker.io/docker/dockerfile:1
#2 DONE 2.7s

#3 docker-image://docker.io/docker/dockerfile:1@sha256:4a43a54dd1fedceb30ba47e76cfcf2b47304f4161c0caeac2db1c61804ea3c91
#3 CACHED

#4 [internal] load metadata for docker.io/library/python:3.10-alpine
#4 DONE 1.3s

#5 [internal] load .dockerignore
#5 transferring context:
#5 transferring context: 2B done
#5 DONE 0.2s
```

```
superxoma@superxoma:~/devops11kube/kustomize$ kubectl apply -k dev
service/dev-redis unchanged
service/dev-service-devops unchanged
deployment.apps/dev-flask-app configured
deployment.apps/dev-redis unchanged
superxoma@superxoma:~/devops11kube/kustomize$ kubectl apply -k prod
service/prod-redis unchanged
service/prod-service-devops unchanged
deployment.apps/prod-flask-app configured
deployment.apps/prod-redis unchanged
```

```
superxoma@superxoma:~/devops11kube/kustomize/base/flask_redis$ minikube ssh
Linux minikube 6.8.0-101-generic #101-Ubuntu SMP PREEMPT_DYNAMIC Mon Feb  9 10:15:05 UTC 2026 x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
docker@minikube:~$ curl 10.98.226.57:54321
Hello World! I have been seen 1 times. My name is: dev-flask-app-684568b78d-hkrbm My env: DEVELOPMENT
```

```
docker@minikube:~$ curl 10.98.226.57:54321
Hello World! I have been seen 3 times. My name is: dev-flask-app-684568b78d-hkrbm My env: DEVELOPMENT
docker@minikube:~$ curl 10.96.29.143:12345
Hello World! I have been seen 3 times. My name is: prod-flask-app-76574c94b-hk2m5 My env: PRODUCTION
```